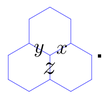
Exact Organization of Density Matrices and Entanglement Structure in the Kitaev Spin Liquid

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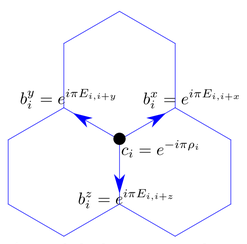
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We give an exact form of the density matrix **(DM)** of the spin-1/2 Kitaev spin liquid represented in terms of spin operators and study the entanglement properties of the Kitaev model within the spin framework. With the explicit form of the DM, plus the exact Gauss law of the emergent gauge theory and the exact 1-form Wilson symmetry in the Kitaev model, we demonstrate the existence of the underlying block-diagonal structure of the reduced density matrix **(RDM)**, which gives rise to the extensive degeneracy in the entanglement spectrum **(ES)**. The block-diagonal structure is then proven to be responsible for the separability of the entanglement entropy **(EE)** into the gauge and matter parts. Furthermore, the method for applying our theory to cases with an odd number of lattice sites is also discussed. It also demonstrates a relation between the ES and the fermion parity **(FP)**, which is seldom mentioned in the literature.

Introduction

- The Kitaev model $H = -\sum_{\langle ij\rangle} J_{\alpha_{ij}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}}$; $\alpha_{ij} = x, y, z$ 

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- The Majorana-fermion representation of spins enlarges the Hilbert space: $\sigma^\mu \to i b^\mu c$. After the transformation, the Hamiltonian becomes $H = i\sum_{\langle ij\rangle} J_{\alpha_{ij}} c_i \hat{u}_{ij} c_j$, where $\hat{u}_{ij} \equiv i b_i^{\alpha_{ij}} b_j^{\alpha_{ij}}$.
- Short-range two-point correlation functions: $\langle \sigma_i^\mu \sigma_j^\nu \rangle = 0$ if (i, j) is not a pair of nearest-neighbor sites or $\mu, \nu \neq \alpha_{ij}$.
- $\mathbb{Z}_2$ gauge theory: $\hat{u}_{ij} = e^{iA_{ij}}$. The b-Majorana fermions serve as the electric field: $b_i^\mu = e^{i\pi E_{i,i+\mu}}$, $b_i^\mu \hat{u}_{i,i+\mu} b_i^\mu = i\hat{u}_{i,i+\mu} \Leftrightarrow [E_{ij}, A_{i'j'}] = -i\delta_{ii'}\delta_{jj'}$.
- $\mathbb{Z}_2$ gauge flux and Stokes law: $e^{i\Phi} = W_p \equiv \prod_{\langle ij\rangle \in p} \hat{u}_{ij} = e^{i \oint A}$
- The Gauss law is $b_i^x b_i^y b_i^z c_i = 1$, which reduces to $\boldsymbol{\nabla} \cdot E = \rho$ if we let $c_i = e^{-i\pi\rho_i}$ be the matter field.



- Tracing out the environment gives an RDM $\rho_A \equiv \mathrm{Tr}_{\overline{A}}\rho$ of a region A, which encodes the bipartite entanglement. It is an effective local state: $\langle \mathcal{O}_A \rangle = \mathrm{Tr}\rho\mathcal{O}_A = \mathrm{Tr}_A \rho_A \mathcal{O}_A$.

Motivation

- An exact form of **DM** and **RDM** of a quantum spin liquid **(QSL)** state represented in terms of **spin**, which may pave the way for research on quantum resource theory.
- Clarify the physical meaning of **entanglement from the gauge field**.
- A direct demonstration of the **ES** of the Kitaev ground state with **spin representation** without relying on the **replica trick** [2]
- Investigate how to deal with **odd cases** and demonstrate the interplay between subsystem **FP** and **ES**.

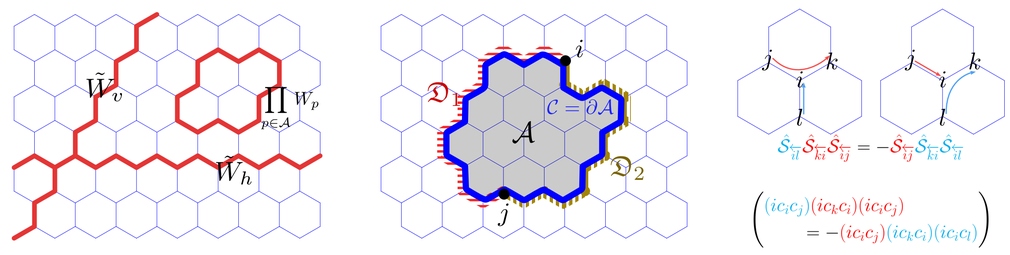
Correspondence between Strings and Fermions

- String operators and plaquette projectors as the building blocks:

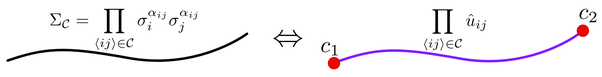
$$\hat{\Sigma}_{\mathfrak{D}} \equiv \prod_{\langle ij\rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}}, \qquad \hat{\Pi}_A \equiv \prod_{p \in A} \left(\frac{\mathbb{1} + W_p}{2}\right)$$

At the ground state (flux-free sector), the strings are projective: $\mathcal{S}_{\mathfrak{D}} \equiv \hat{\Pi}\hat{\Sigma}_{\mathfrak{D}}$

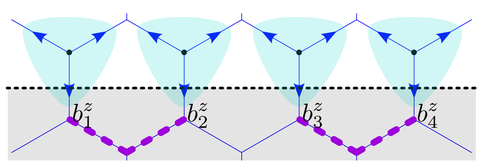
- **Equivalence class**: $\mathfrak{D} \in [\mathfrak{D}^c]$ if $\mathfrak{D} = \mathfrak{D}^c \ominus \partial\mathcal{A}$, with or without global loops. An equivalence class only contains information about **endpoints** $\mathrm{EP}(\mathfrak{D})$.



- Strings and the endpoints: Fermions emerge at the endpoints of string operators, and string braiding reproduces their statistics.



- Gauge-field entanglement from Gauss law [3]: Each *residual* Gauss law gives an operator that commutes with the RDM, and contributes a $\frac{1}{2}\ln 2$ to the EE. The problem is that **the residue operators, b^μ's, are not physical**. The solution to this problem is **linking them via strings**.



$$\hat{\boldsymbol{\xi}} \equiv \sigma_1^{\mu_1} \hat{\Sigma}_{\mathcal{C}} \sigma_2^{\mu_2} \propto b_1^{\mu_1} \left(\prod_{\langle ij\rangle\in\mathcal{C}} \hat{u}_{ij}\right) b_2^{\mu_2}$$

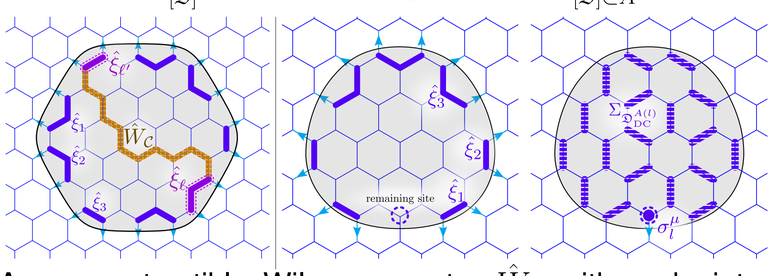
Results

- A DM of a spin-1/2 system can be expanded by Pauli strings:

$$\rho = \frac{1}{2^N} \sum_{\{\mu_i = 0,x,y,z\}} \langle \sigma_1^{\mu_1} \cdots \sigma_N^{\mu_N} \rangle \sigma_1^{\mu_1} \cdots \sigma_N^{\mu_N}$$

- Short-range two-point correlation: The surviving Pauli strings must be string operators $\hat{\Sigma}_{\mathfrak{D}}$.
- DM and RDM of the Kitaev ground state:

$$\rho = \frac{1}{2^{N/2}} \sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}} \hat{\mathcal{S}}_{\mathfrak{D}}, \quad \rho_A = \frac{1}{2^{\frac{N^A}{2} + \left(\frac{1}{2}|\partial A| - 1\right)}} \sum_{[\mathfrak{D}]\subset A} \mathfrak{C}_{\mathfrak{D}} \hat{\mathcal{S}}_{\mathfrak{D}}^{(A)}$$

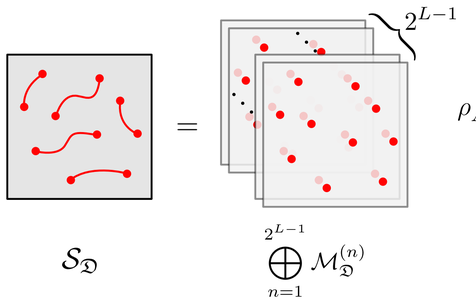


- A non-contractible Wilson operator $\hat{W}_C$ with endpoints in contact with the boundary flips ξ-operators in contact with their endpoints [3]. Therefore, they relate blocks in the following way

$$\hat{W}_C \varrho^{(A)}_{\{\cdots,\xi_i,\cdots,\xi_j,\cdots\}} \hat{W}_C = \varrho^{(A)}_{\{\cdots,-\xi_i,\cdots,-\xi_j,\cdots\}}$$

where i and j are connected to the endpoints of $\hat{\mathcal{W}}$.

- The residual Gauss law defines $\hat{\xi}_i \equiv \sigma_{i_1}^{\mu_{i_1}} \hat{\Sigma}_{\mathcal{C}_i} \sigma_{i_2}^{\mu_{i_2}}$, which block-diagonalize the RDM.



$$\rho_A = \frac{1}{d} \bigoplus_{\{\xi\}} \varrho^{(A)}_{\{\xi\}} = \frac{1}{2^{(|\partial A|/2-1)}} \bigoplus_{n=1}^{2^{(|\partial A|/2-1)}} \varrho_A^{(n)}$$

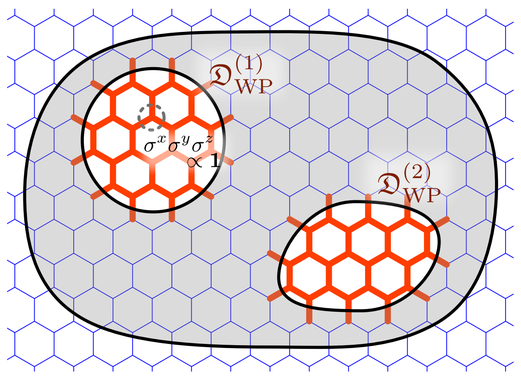
- For $N^A \in$ **odd**, an operator Ξ_A composed of dimer covering and a singular spin operator commutes with the RDM, leading to an additional 2-fold degeneracy.

$$\Xi_A|_{\forall u=1} = \sigma_l^\mu \Sigma_{\mathfrak{D}_{\mathrm{DC}}^{A(l)}} \propto b_i^\mu \prod_{j\in A, j\neq i} c_j$$

- The Majorana representation of Ξ_A operator reflects its physical meaning: the subsystem **FP**.

Outlook

1. Multiply-connected regions: $\mathrm{Tr}_{\overline{A}}\Sigma_{\mathfrak{D}_{\mathrm{WP}}} \neq 0$.



2. Chiral spin liquid in the Yao-Kivelson model [4]
3. String-gas tensor product operator [5]
4. **Quantum magic** (Stabilizer Rényi entropy)

$$\tilde{M}_2(\rho_A) = -\ln\left(\frac{\sum_{[\mathfrak{D}]\subset A} \mathfrak{C}_{\mathfrak{D}}^4}{\sum_{[\mathfrak{D}]\subset A} \mathfrak{C}_{\mathfrak{D}}^2}\right)$$

5. Finite-temperature physics and open quantum systems (Lindbladian dynamics) [6]

Conclusions

- The DM of the Kitaev ground state, expressed in terms of spin operators, is obtained, and the ES is thoroughly investigated within this framework.
- The **exact Wilson 1-form symmetry** sorts string configurations into **equivalence classes** according to their **endpoints**, implying that **Majorana fermions** emerge at the endpoints of strings.
- The degeneracy in the ES implies that the EE contributed by the Gauss law is $S_A = \left(\frac{1}{2}|\partial A| - 1\right)\ln 2$. No need for the replica trick.
- The ξ-operators produced by the residual Gauss law block-diagonalize the RDM. Each block is the DM of the equivalent Majorana fermion system.
- All blocks are identical because the non-contractible Wilson operator relates every block. The ES is thus that of the equivalent Majorana system with **degree of degeneracy** $2^{\frac{1}{2}|\partial A|-1}$, which contributes to the gauge-field entanglement.
- For regions with an odd number of sites, an operator Ξ_A can be found to further block-diagonalize the RDM, marking the intrinsic difference between even and odd **subsystem FP**.
- The Ξ-operator illustrates that the matter sector borrows a fermion from the gauge sector to make the FP well-defined. This is valid only in the spin formulation, in which the matter and gauge sectors are treated as a whole.

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